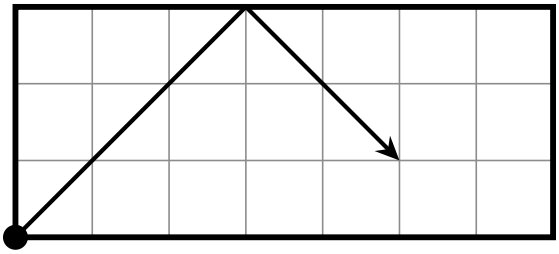
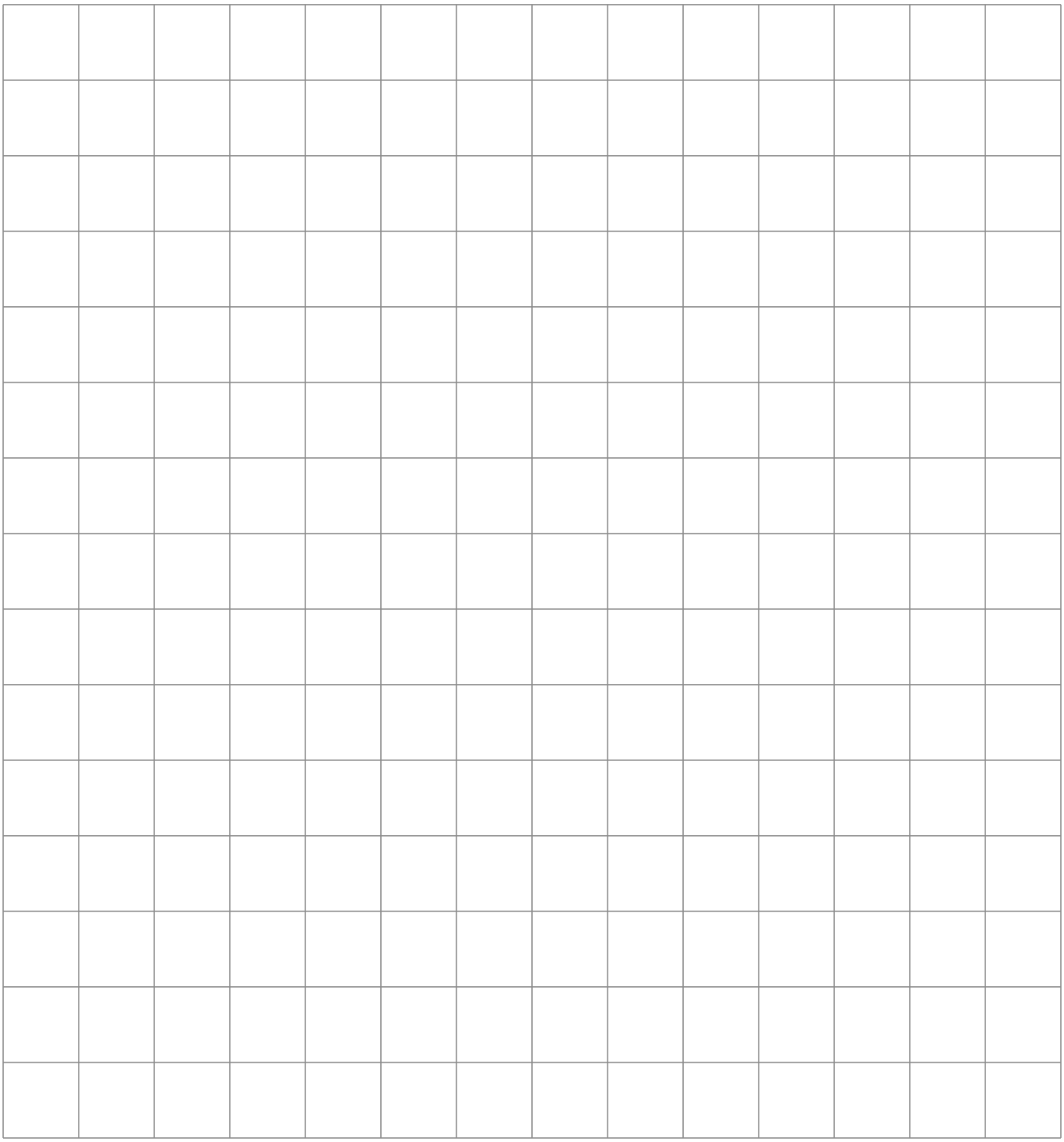
The ball starts at the dot in the bottom left corner of a table and rolls up and to the right, crossing each square from corner to corner. It bounces off every wall it hits and stops when it reaches a corner of the table. Each time the ball hits a wall counts as one bounce; reaching the corner at the end does not count. A 7 by 3 table is 7 squares wide and 3 squares high.

Partners take turns: one predicts where the ball will stop, and the other traces the path.



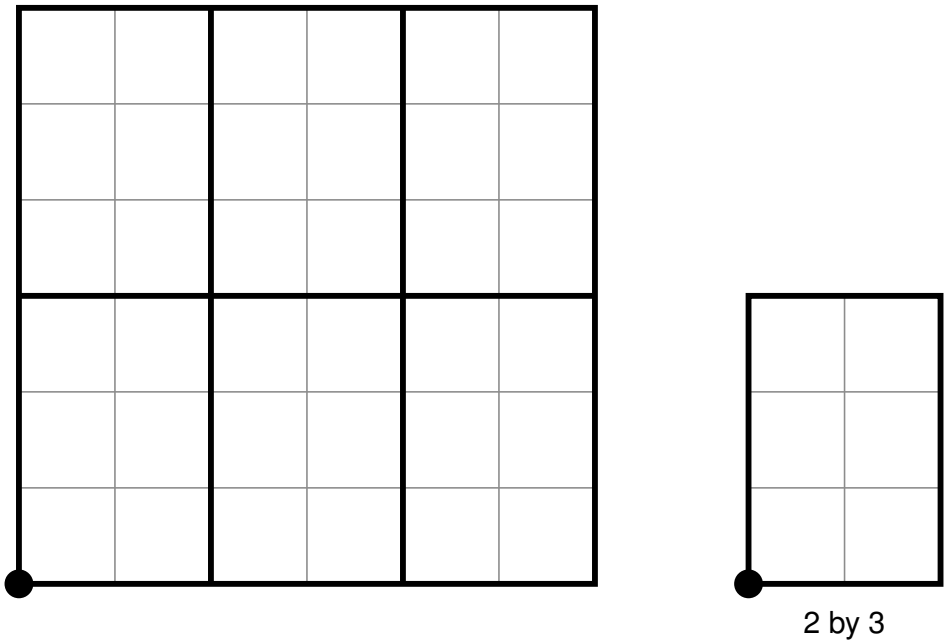
7 by 3

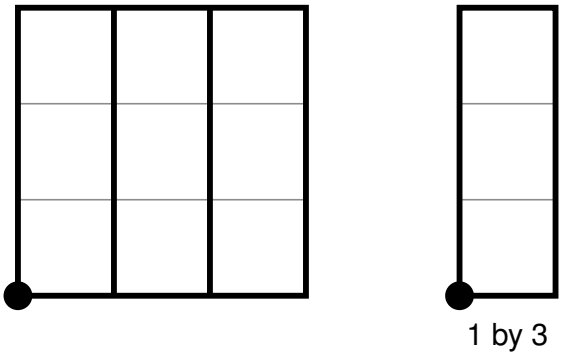
Problem 5: Play this game with your partner. One player draws a table on the grid, no more than 6 squares wide and 6 squares high, and marks its dot. The other player says where the ball will stop and how many times it will bounce, and then traces the path. The player who predicted scores 1 point for the right corner and 1 point for the right number of bounces. Then swap jobs. After each player has predicted three tables, the player with more points wins.



Problem 6: Find every table no more than 6 squares wide and no more than 6 squares high where the ball bounces exactly once. Explain how you know you have found them all.

Problem 7: With a ruler and a dark marker, draw a straight line from the dot to the opposite corner of each sheet. How many thick lines inside the sheet does each line cross? Then cut out the tables each line passes through, keeping them in one piece. Fold them on the thick lines until they are the size of one table with the dot in front. Hold them up to the light, and copy the path you see onto the small table beside the sheet.





Problem 8: Is there a table where the ball stops at the corner it started from? Find one, or explain why there is none.